THE FORBIDDEN INTERVAL

An Experiment in Algorithmic Composition

Concept & Functional Specification

Sonic Pi Generative Composition | Version 1.0

Central idea

A probabilistic composition engine in which the augmented fourth (tritone) triggers a change from C Dorian to F-sharp Locrian, while a perfect fifth provides the return mechanism.

Prepared as a documentation and presentation source. The text is designed to be reusable in an explanatory YouTube video and may be revised as the project develops.

1. Project Overview

THE FORBIDDEN INTERVAL is a generative musical composition developed in Sonic Pi. It explores the relationship between two modal environments—C Dorian and F-sharp Locrian—using the augmented fourth, historically associated in Western musical discourse with *Diabolus in Musica* ("the Devil in Music"), as a structural trigger.

Rather than following a fixed melody or predetermined arrangement, the system generates notes through random selection and changes its musical state according to probability checks and minimum event-count requirements.

The central concept is therefore not simply to alternate between two scales. It is to use a distinctive interval as an event that changes the rules governing subsequent note generation.

1.1 Core Musical Cycle

C Dorian → F-sharp Locrian → C Dorian → ...

Event	Function	Pitch relationship
Augmented fourth	Triggers Dorian-to-Locrian transition	C to F-sharp: +6 semitones
Perfect fifth	Triggers Locrian-to-Dorian return	F-sharp to C-sharp: +7 semitones

The two events are intentionally asymmetric: they use different minimum event counts, probabilities, and pause lengths. This contributes to the changing duration and character of each state.

2. Musical Architecture

2.1 Default State: C Dorian

The composition starts in C Dorian. Its interval pattern, measured from the tonic, is:

```
0, 2, 3, 5, 7, 9, 10
```

Notes: C – D – E-flat – F – G – A – B-flat.

While the system is in this state, it randomly selects notes from the C Dorian scale. A counter records the number of successfully generated Dorian notes. Once the minimum count has been reached, the system becomes eligible to transition to Locrian.

2.2 Alternative State: F-sharp Locrian

The Locrian root is calculated from the Dorian root by adding six semitones. With C as the starting root (MIDI note 60), the result is F-sharp (MIDI note 66).

```
0, 1, 3, 5, 6, 8, 10
```

Notes: F-sharp – G – A – B – C – D – E.

While the system is in this state, it randomly selects notes from F-sharp Locrian and increments a separate Locrian event counter. Once the minimum count has been reached, the system becomes eligible to return to Dorian.

3. The Forbidden Interval: Dorian-to-Locrian Transition

The augmented fourth is the central structural mechanism. It is calculated as the Dorian root plus six semitones. In this implementation, the resulting pitch is F-sharp.

A transition can occur only when both of the following conditions are met:

• At least five Dorian notes have been generated.
• A random probability check succeeds, using a configured probability of 0.05 (5%).

When the conditions are met, the system plays the F-sharp transition note, changes its state to Locrian, resets the Locrian event counter, and inserts a two-beat pause. The transition note uses the same synthesizer as the rest of the

composition, but has a longer release and a slightly reduced amplitude.

3.1 Musical Significance

F-sharp is outside the C Dorian scale. Its appearance therefore marks a departure from the established modal vocabulary. The system does not simply add a chromatic note and continue as before: it changes the scale used for subsequent note selection.

The tritone consequently functions as more than a dissonant interval. It becomes a structural event that transforms the musical environment.

4. Return Mechanism: Locrian-to-Dorian Transition

The return event is calculated as the Locrian root plus seven semitones. For F-sharp, this produces C-sharp.

A return can occur only when both conditions are met:

• At least three Locrian notes have been generated.
• A random probability check succeeds, using a configured probability of 0.15 (15%).

When the conditions are met, the system plays C-sharp, changes its state back to Dorian, resets the Dorian event counter, and inserts a three-beat pause.

4.1 An Important Musical Distinction

C-sharp is a perfect fifth above F-sharp, but it is not the tonic of C Dorian. The code therefore implements a state transition rather than a conventional dominant-to-tonic resolution to C. The return is defined by the relationship between the Locrian root and the transition note, not by a classical cadence back to the original tonic.

Characteristic	Dorian → Locrian	Locrian → Dorian
Trigger pitch	F-sharp	C-sharp
Interval	Augmented fourth (+6)	Perfect fifth (+7)
Probability	5%	15%
Minimum generated notes	5	3
Additional pause	2 beats	3 beats

5. Generative Sequencing

5.1 Random Note Generation

Each sequencing iteration has a 35% probability of generating a note. If a note is generated, it is selected randomly from the scale associated with the current state. There is no predefined melody or fixed rhythmic sequence.

5.2 Timing and Silence

Between iterations, the system waits for a randomly selected duration between three and twelve beats. At 60 BPM, this corresponds to approximately three to twelve seconds. Silence is therefore an intentional part of the musical texture.

The transition events introduce additional pauses: two beats after the augmented-fourth transition and three beats after the perfect-fifth return.

5.3 Event Counters and Eligibility

The Dorian and Locrian counters increment only when an ordinary note is successfully generated in the corresponding state. They do not count loop iterations or elapsed time. Each counter is reset when the system enters its associated state.

Implementation detail: The transition probability is evaluated on every loop iteration after the minimum event count has been reached—not only on iterations in which an ordinary note is generated. Thus, the 5% and 15% values are probabilities per eligible loop iteration, not per generated note.

6. Sound Design

6.1 Synthesizer and Effects

The same Sonic Pi synthesizer, **:pretty_bell**, is used for ordinary notes and transition events. A global reverb effect is applied throughout the composition.

Parameter	Value
Synthesizer	:pretty_bell
Tempo	60 BPM
Reverb room	0.9
Reverb mix	0.45
Base amplitude	0.5

6.2 State-Dependent Sound Parameters

Event	Amplitude	Release	Pan
Dorian note	0.50	3 beats	–0.15 to +0.15
Locrian note	0.425	4 beats	–0.25 to +0.25
Augmented-fourth transition	0.40	5 beats	Center
Perfect-fifth return	0.50	5 beats	Center

Locrian notes are slightly quieter and have a longer release than Dorian notes. The Locrian state also uses a somewhat wider stereo-pan range. Both transition events are centered in the stereo field.

7. Configuration Parameters

Parameter	Default	Description
dorian_root	60	MIDI note number for C; basis for transposition.
transition_probability	0.05	Probability of entering Locrian on an eligible loop iteration.
resolution_probability	0.15	Probability of returning to Dorian on an eligible loop iteration.
min_dorian_events	5	Minimum generated Dorian notes before transition eligibility.
min_locrian_events	3	Minimum generated Locrian notes before return eligibility.
note_density	0.35	Probability of generating an ordinary note per loop iteration.
rest_min	3	Minimum inter-iteration rest, in beats.
rest_max	12	Maximum inter-iteration rest, in beats.
base_amp	0.5	Base note amplitude.
BPM	60	Global tempo.

The scale definitions and transition pitches are derived from the Dorian root. Changing the root therefore transposes the system while preserving the six-semitone relationship between the two modal roots.

8. State-Transition Logic

The system runs continuously in one of two states. In Dorian, it generates notes from C Dorian and checks whether the augmented-fourth transition conditions have been met. If the transition succeeds, it plays F-sharp and enters Locrian. In Locrian, it generates notes from F-sharp Locrian and checks whether the perfect-fifth return conditions have been met. If the return succeeds, it plays C-sharp and re-enters Dorian.

There is no predetermined ending, fixed duration, or fixed number of transitions. The composition continues until playback is stopped.

9. Intended Musical Characteristics

Modal stability. C Dorian establishes a recognizable initial environment. Random note selection creates variation while retaining a defined scale vocabulary.

Unexpected destabilization. The F-sharp transition note lies outside C Dorian and marks a change in the governing musical rules.

Contrasting modal environment. F-sharp Locrian supplies a different pitch collection rather than merely adding occasional chromatic notes to Dorian.

Probabilistic return. The C-sharp event initiates a return to the Dorian state without functioning as a conventional resolution to C.

Nonlinear development. The music evolves through the interaction of probability, pitch selection, timing, silence, and state transitions.

10. Composer’s Listening Evaluation

10.1 Overall Impression

After listening to the actual output generated by this Sonic Pi implementation, the composer found the result remarkably compelling and musically engaging. The music is more than a technical demonstration of a generative algorithm: it has a distinctive character and sustains interest as a listening experience in its own right.

The combination of randomized note selection, variable silence, modal changes, and probabilistic transitions produces an evolving musical landscape. Although the underlying rules are relatively compact, their interaction creates results that feel varied and difficult to predict.

10.2 Structure and Unpredictability

A particularly successful aspect is the balance between constraint and randomness. The system is not arbitrary: each state has a defined modal vocabulary, and each transition is governed by specific conditions. Yet the timing of individual notes and changes of state is not predetermined.

This balance allows the listener to sense an underlying musical identity without knowing exactly what will happen next. The minimum event counts provide a degree of continuity, while probability and randomized timing prevent the composition from settling into a fixed pattern.

10.3 The Tritone as a Compositional Event

The most distinctive idea is the use of the augmented fourth as a structural trigger. Its appearance does not merely add dissonance to an otherwise unchanged passage. It changes the scale from which subsequent notes are selected, transforming the musical environment itself.

The historical association of the tritone with tension is thus translated into an algorithmic compositional principle: an interval becomes an agent of musical change.

10.4 Composer's Assessment

The listening experience confirms that the concept works musically. The generated music is unexpectedly engaging, and the relationship between modal stability, disruption, and return gives the system a compelling sense of development. The project's success lies not only in the fact that the code behaves as designed, but in the musical quality of the result.

The composer's assessment is that this implementation should be preserved as a successful original version before further experimentation. Future modifications should be treated as separate iterations so that the musical character of Version 1.0 remains available for comparison.

11. Compositional Philosophy and Conclusion

THE FORBIDDEN INTERVAL explores the possibility of turning a historically significant musical interval into a mechanism for generative composition. The tritone is treated not merely as a sound, but as an event that changes the system's behavior. The perfect fifth provides a complementary, though not conventionally tonal, return mechanism.

The composition therefore revolves around an ongoing relationship between stability, disruption, transformation, and return. Rather than specifying every note in advance, the composer defines the conditions under which the music can evolve.

> **The composition is not a predetermined piece of music. It is a system capable of generating an evolving musical experience.**

Appendix A. Key Sonic Pi Logic

The following excerpts summarize the pitch relationships at the heart of the implementation:

```
locrian_root = dorian_root + 6

augmented_fourth = locrian_root
perfect_fifth = locrian_root + 7

# C (60) + 6 = F-sharp (66)
# F-sharp (66) + 7 = C-sharp (73)
```

Appendix B. Version 1.0 Scope Notes

This document describes the behavior of the supplied Sonic Pi code. It does not assume additional features such as a simultaneous C–F-sharp dyad, a dedicated visualizer, live parameter controls, or a separate audio layer. Those would be potential extensions, not features of the current implementation.

The transition pitch F-sharp is played as a single note; the code does not sound C and F-sharp together as a literal tritone chord or dyad. The term "tritone transition" refers to the six-semitone relationship between the original Dorian root and the new Locrian root.